DEEP RESEARCH REPORT

Project AURUM

A Jev-inspired looped-transformer decision model for gold (XAUUSD): findings, limitations and build plan

Prepared for: tskdev (Kaggle account)

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Method: DeerFlow deep-research pipeline - 5 parallel threads, 180+ sources

Stack: Kaggle free tier / Hugging Face / Dukascopy / OANDA practice

Executive Summary

This report answers one question: can a solo builder, on a strictly free technology stack, construct a Jev-inspired “System One” trading decision model for gold (XAUUSD) whose edge survives honest statistical validation? The research was commissioned after the project direction was fixed: take an open pretrained transformer of roughly 0.6B parameters, convert it into a looped (recurrent-depth) architecture, and train it to output typed, calibrated trading decisions - entry, exit, risk, and position size - with all compute on Kaggle's free GPUs and all data from free sources. Five parallel research threads, run under the DeerFlow deep-research methodology, examined more than 180 primary sources between architecture science, the Jev reference model, AI-trading evidence, market microstructure, and statistical validation.

Five findings drive the recommendation. First, TypeSafe AI's Jev is real, fast and closed: its weights, architecture and training method (RLCD) are unpublished, so replicating its behavior requires building our own - but its published failure modes and one verified 17-day real-money experiment give us an unusually precise design checklist. Second, the looped-transformer science base has matured exactly where we need it: LoopUS (May 2026) demonstrates a recipe for converting a stock Qwen3-class model into a looped refiner with about 3B training tokens, and independent 2026 results show looping helps most on structured and time-series data and when data is the binding constraint - precisely our situation. Third, the market context is severe: gold sits near $4,053 after a 27% drawdown from its January 2026 peak of $5,586, with realized volatility near 30% annualized, so the backtest window contains both a parabolic rally and a crash - an honest stress test most projects never face. Fourth, the verified performance ceiling for individuals is a net-of-cost Sharpe ratio between roughly 0.5 and 1.0; every claim materially above that range, in the academic and practitioner literature alike, failed structural validity checks. Fifth, no published work applies looped transformers to trading - the niche is open, which is both an opportunity and a warning that no precedent validates the idea.

The report therefore recommends building AURUM-0.6B: Qwen3-0.6B-Base converted with a scaled-down LoopUS surgery (8 loop steps, 3 supervised depths), trained in three stages - looped conversion, decision training with a calibration-penalized loss, then weekly online learning - and wrapped in a deterministic risk engine: the model judges, code executes. Validation follows the six-point Alpha Illusion protocol with combinatorial purged cross-validation and Deflated Sharpe Ratio gates against mandatory LightGBM and GRU baselines. The 16-week phased plan runs on Kaggle's free GPU quota and ends in extended paper trading on a free practice account. The honest target is a net Sharpe of 0.5 to 1.0; the final chapter enumerates twelve ways this plan can fail, each with its mitigation.

Table of Contents

Executive Summary....1
1. Background, Objectives and Scope....1
2. Research Methodology (DeerFlow Pipeline)....2
3. The Reference Model: Jev and System One Models....2
3.1 What Jev is - and is not....3
3.2 Nine published failure modes....3
3.3 The 17-day real-money experiment....3
3.4 The fast-decision landscape, late 2026....4
3.5 Eight design lessons carried into the architecture....4
4. Architecture Science: Looped and Recurrent-Depth Transformers....5
4.1 The mechanism....5
4.2 The four model families that matter....5
4.3 The skeptical evidence - what looping does not do....6
4.4 Why the evidence favors looping for this project....6
5. AI Trading Systems: What the Evidence Says Works and Fails....6
5.1 The flagship claims do not survive scrutiny....7
5.2 The evaluation protocol we adopt: Alpha Illusion, P1-P6....7
5.3 Foundation models for time series are not a free lunch....7
5.4 The one differentiator that keeps appearing: online learning....7
6. The Best Results Ever Achieved by Individuals....8
6.1 Kaggle finance competitions....8
6.2 Numerai, and the honest professionals....8
6.3 The failure statistics....8
7. Market Reality: Gold in November 2026....9
7.1 Where the market stands....9
7.2 The full regime map....10

7.3 Costs, sessions and events - the numbers the backtest must charge....10
8. Component Selection: Base Model, Data and Backtesting Stack....10
8.1 Base model....10
8.2 Data....11
8.3 Backtesting stack....11
9. Free-Stack Training Feasibility on Kaggle....11
9.1 What the free tier actually provides (November 2026)....11
9.2 Throughput math, and the one big adjustment....12
10. Validation Protocol and Risk Management Framework....12
10.1 The statistics that kill backtests....13
10.2 Data splits, contamination controls and calibration....13
10.3 Risk management: sized for the edge we can prove....13
11. Master Limitations Table and Mitigations....14
12. Recommended Architecture and Phased Build Plan....14
12.1 The AURUM-0.6B system....14
12.2 The seven phases....15
12.3 What success looks like - honestly....15
13. Key Sources and Confidence Notes....15
13.1 Architecture (looped transformers)....15
13.2 Jev, System One models and AI-trading evidence....16
13.3 Individuals, statistics and risk....17
13.4 Market data and infrastructure....17
13.5 Low-confidence items (flagged, not relied upon)....18

Note: This Table of Contents is generated via field codes. To ensure page number accuracy after editing, please right-click the TOC and select "Update Field."

1. Background, Objectives and Scope

The project brief, fixed before research began, is deliberately ambitious: build an AI trading decision model for gold and foreign exchange, modeled on TypeSafe AI's Jev - a “System One” model that reads a market state and returns typed, probabilistic decisions in a fraction of a second - but implemented as an open, self-hosted looped transformer rather than a proprietary API. The backbone must be an existing pretrained model in the 0.6B-parameter class, with its transformer converted to a recurrent-depth (looped) architecture. The system must own the full decision stack: direction (long, short or flat), entry and exit timing, stop placement, position sizing, and portfolio-level risk, all evaluated through backtesting. Two constraints are absolute: the entire stack must be free (Kaggle compute, open weights, free data, free paper trading), and the process must be automated end to end through the Kaggle API, which was verified working before this research began.

The user's stated research philosophy sets the method: “research end to end first, plan second, build last,” with first priority given to the best results ever verified from individual builders rather than institutional claims, and with explicit instructions to find the limitations of the plan and fix them before writing code. This report is the output of that research phase. It deliberately includes negative findings - inflated benchmark claims, documented failure rates of retail algorithmic trading, and the statistical traps that make most backtests worthless - because those findings define the engineering requirements as concretely as the positive ones.

Scope is bounded as follows. The trading instrument is spot gold (XAUUSD) against the US dollar, chosen by the user, with possible later expansion to major FX pairs. The decision cadence is hourly or slower, on holding periods from several hours to several days; the research in Chapter 8 explains why faster horizons are structurally unprofitable at retail cost levels. The deliverable of the build phase is a trained model plus a deterministic execution and risk layer, validated in backtest and paper trading. This report is research and engineering documentation, not investment advice, and it promises no returns; Chapter 11 quantifies exactly how uncertain any backtest result is at achievable sample sizes.

2. Research Methodology (DeerFlow Pipeline)

The research followed the deep-research methodology of ByteDance's open-source DeerFlow framework, whose repository (github.com/bytedance/deer-flow) was cloned and studied at the start of this exercise. DeerFlow's official deep-research skill prescribes four phases: broad exploration to map the territory, targeted deep dives per dimension with multiple query phrasings, diversity and validation across facts, examples, expert opinions

and criticisms, and a final synthesis check. Two of its rules shaped this work most: never generate content from a single search, and actively seek challenges and limitations rather than only supporting evidence. Because the DeerFlow harness itself requires external LLM and search API keys that would violate the free-stack constraint, the pipeline was executed natively with equivalent tooling: five parallel research agents, each assigned one thread, each instructed to read primary sources in full and to tag every finding with a confidence level.

Table 1: The five parallel research threads

Thread	Question answered	Representative sources
A. Architecture	Looped / recurrent-depth transformers: mechanisms, conversion recipes, public checkpoints, criticisms	arXiv papers (Huginn, LoopUS, Ouro, ARLoop); Hugging Face model cards; GitHub repos
B. Reference model and trading AI	What Jev actually is; LLM trading frameworks; best verified individual results	typesafe.ai blog and docs; Wikipedia; Kaggle leaderboards and writeups; Numerai; arXiv
C. Base model and compute	Base model selection; tokenizer digit handling; Kaggle free-tier limits; training throughput	Hugging Face configs (tested empirically); Kaggle docs; Unsloth benchmarks
D. Data and backtesting	Free XAUUSD data depth and limits; backtest frameworks; transaction costs; horizon evidence	Live API tests (Yahoo, dukascopy-node); PyPI release data; broker cost sheets; academic studies
E. Limitations and risk	Overfitting statistics; regime history; contamination; risk of ruin; calibration	Bailey and Lopez de Prado papers; FCA/AMF documents; World Gold Council; FRED; computed math

Findings are tagged HIGH (primary source read in full during this research), MEDIUM (secondary source or solid multi-source agreement) or LOW (single unverified claim). Cross-checks between threads resolved most discrepancies; the few that remain are listed in Chapter 14. One example: the 17-day Jev trading experiment was read in full directly (HIGH), although a later search thread could not re-locate the article through its index - the direct reading stands. Roughly 180 sources were examined; the most load-bearing forty are cited in the chapters and listed at the end.

3. The Reference Model: Jev and System One Models

3.1 What Jev is - and is not

Jev, released on 15 September 2026 by San Francisco startup TypeSafe AI, is a hosted decision model the company calls a “System One model,” after Kahneman's fast, intuitive

mode of thinking. It is not a chat model and does not generate text. Its API takes a state - unstructured text or structured JSON context - plus typed questions, and returns, in 70 to 500 milliseconds, a calibrated probability for each answer: choice questions return a probability per option, score questions return a distribution over ordered levels, and noul questions return the probability that a statement is true. Questions about the same state are evaluated in parallel in a single pass. Input costs $0.042 per million tokens and output is free, which the company quantifies as 40 to 200 times faster and far cheaper than comparable LLM calls on decision-shaped queries. The company, founded in 2024 by ex-OpenAI researcher Diogo Almeida and colleagues, raised a $40M seed round led by DCVM and was reported in October 2026 to have closed a much larger round at a multi-billion valuation (MEDIUM confidence).

Critically for this project: Jev's weights are closed. TypeSafe publishes no architecture details beyond describing it as transformer-based, offers no fine-tuning, and its training method - “Reinforcement Learning for Calibrated Decisions” (RLCD) - has no published paper. Wikipedia's editors note outside speculation that Jev is trained on synthetic data with a loss that penalizes miscalibration, possibly a Brier-score term (MEDIUM). An attempted open replication by Together AI (“tev1”, a Qwen3.5-4B fine-tune on about 38k labeled questions) matched single-label speed but did not reproduce calibrated confidence - and the community's own comparison site concludes that across 54 open clones, “calibrated confidence is the part nobody has matched in public.” Calibration, not raw accuracy, is Jev's actual moat, and therefore the core training objective our replica must get right.

3.2 Nine published failure modes

TypeSafe's own documentation lists nine failure modes for Jev 1.13, independently summarized by the community. They matter because each one becomes a design requirement for our system.

Table 2: Jev 1.13 documented failure modes and design consequences for AURUM

Failure mode (published)	Design consequence for our build
Reads instructions literally; boundary cases unreliable	Questions must be atomic and unambiguous; two literal questions beat one interpretive one
Maths and counting unreliable	All arithmetic (returns, ATR, sizing) computed in code, never by the model
Dates unreliable	Time features pre-computed and encoded numerically
Indirection fails (“judge the field the user asked about”)	Each question names its target explicitly
Large state: irrelevant detail lowers accuracy	State kept minimal: only pre-filtered, task-relevant features

Failure mode (published)	Design consequence for our build
Prompt injection from state content	Market data is numeric and pre-binned - no free text from third parties enters the state
Contradictory instructions confuse it	One decision policy, encoded once, in the execution layer
Option-order bias: leans toward the first option	Average decisions over randomized option orders, or use symmetric heads
Cannot generate; extract candidates elsewhere	Our model only scores fixed actions - it never generates text

3.3 The 17-day real-money experiment

The most valuable single document found in this research is a verified first-person experiment (Towards AI, 19 October 2026, read in full during this research): a developer ran Jev as the decision layer on a deliberately tiny live account trading crypto perpetual futures on KuCoin, 30 September to 17 October 2026, with 1-to-8-hour holding periods. The design followed the principle “code enforces survival; the model owns opportunity”: Jev answered direction, playbook, entry timing, target distance and position management questions about a state of trend, momentum, funding and stretch indicators; the code held all gates, stops and sizing; the model never saw the balance or its own scoreboard. Questions were asked in every option order and averaged, and every question had a stand-aside option.

The results were poor, and precisely because they were measured, they are instructive: 507 scored calls produced 15 trades, an 8-for-15 (53%) win rate, and a total loss of 4.5R (-1.78% of account). Average win +0.27R against average loss -0.95R; every trade closed on its exchange stop and none reached target. Most damning, confidence ran backwards: calls made at 0.95+ confidence were right 32% of the time, calls under 0.70 confidence were right 57%, and the rank correlation between confidence and outcome was -0.17. Trend-following calls lost consistently in the choppy tape (-0.76% per call), while fading stretched moves was the only profitable behavior (+0.78%). Option ordering moved 26% of answers by 0.10 or more. A parallel LLM agent on the same market lost slightly less (-1.2R), agreed with Jev on 112 of 117 shared calls - and those shared calls were right only 44% of the time, evidence that the two models were echoing each other's priors rather than providing independent judgment. The author's conclusion matches the academic literature in Chapter 6: System One models suit narrow judgements with clear criteria and gates in front of slower reasoning, not raw forecasting of noisy outcomes, and raw confidence must never be used as a threshold or position-size input without independent recalibration.

3.4 The fast-decision landscape, late 2026

Jev now has direct competitors, which confirms the category and gives us benchmarks. OpenAI shipped “Decisions” in public beta on 6 October 2026 (DevDay), powered by GPT-6 Luna, promising sub-second structured decisions. Perplexity offers Decisions at $0.04 per million input tokens. Most important for a free stack: Cloudflare's Clef (clef-flash) undercuts Jev's price at $0.038 per million tokens, claims full Jev-API compatibility - and its 27B model is based on open-weight Qwen3.8-27B, released under Apache-2.0 on Hugging Face, self-hostable on a single H200. Community clones run smaller still: von claims under 15 ms per decision locally, Laya is a 421M-parameter classifier needing about 1 GB of RAM. The open ecosystem, in other words, already believes that decision models can be small; what none of them has replicated is calibrated, per-decision confidence. Our build therefore differs from every clone in exactly one respect that matters: the training objective optimizes decision calibration under regime shift, not classification accuracy.

3.5 Eight design lessons carried into the architecture

- **Model judges, code executes.** The decision model proposes probabilities; a deterministic layer owns stops, sizing, gates and kill switches. This is also the exact architecture the academic literature (Chapter 6) converged on.
- **Never show the model the account state.** Balance, gates and scoreboard stay out of the model state; they are execution-layer facts.
- **Neutralize option-order bias.** Ask symmetric questions or average over randomized option orders; a 26% answer shift from ordering alone is a real, measured failure mode.
- **Give every question an escape.** Stand-aside must always be a first-class action, and in backtests it will likely be the modal action.
- **Do arithmetic in code.** Returns, ATR, volatility and sizing are computed deterministically; the model sees pre-binned features, not raw arithmetic.
- **Keep the state small.** Irrelevant detail measurably degrades decision models; feature count stays disciplined.
- **Treat confidence as a population statistic.** Confidence is calibrated across many answers, not guaranteed per answer; we recalibrate per regime and monitor reliability curves continuously.
- **Expect chop to kill trend-following.** The only profitable behavior in the live experiment was fading stretched moves; regime awareness and mean-reversion behavior must be representable, and evaluated per regime, not on average.

4. Architecture Science: Looped and Recurrent-Depth Transformers

4.1 The mechanism

A looped transformer reuses the same block of layers several times per input, turning depth into a repeatable computation instead of stored parameters. The 2025-2026 literature converged on three stabilizing ingredients. First, a split of the network into an encoder (first layers), a looped block (middle layers, applied K times), and a decoder (final layers) - Huginn's authors call this prelude, core and coda. Second, input re-injection: at every loop iteration the core block receives both the previous latent state and the embedded input, concatenated through a learned adapter, which prevents the state from drifting away from the question being answered. Third, a selective gate (LoopUS) or sandwich normalization (Ouro) that damps each iteration's residual update, so the loop behaves as a convergent refinement rather than a divergent recurrence. At inference, the number of iterations K can be fixed, chosen adaptively by a confidence head, or inferred zero-shot from the divergence between successive iterations - a free uncertainty signal that a decision model can exploit directly. Iterations can be stopped early per decision, and the loop count itself becomes a usable difficulty signal.

4.2 The four model families that matter

Table 3: Looped / recurrent-depth model families relevant to AURUM (all findings from primary papers and model cards)

Family	Approach	Training budget	Key result	Availability
Huginn (Geiping et al., Feb 2025, arXiv 2502.05171)	Pretrained from scratch: (2,4,2) prelude/core/coda, 3.5B total, core loops with input re-injection; KV-cache compression across loops	800B tokens on the Frontier supercomputer (4,096 MI250X GPUs)	Recurrence at r=32 lifts GSM8K from 1.82 to 9.02 vs a matched fixed-depth baseline; effective parameters = K x 1.5B + 2B	Apache-2.0 on Hugging Face (tomg-group-umd/huginn-0125); the rumored 1.5B / “Maxi” checkpoints do not exist
LoopUS (Park et al., May 2026, arXiv 2605.11011)	Post-training conversion of a stock pretrained LLM (Qwen3-1.7B): keep all layers, loop the middle block behind a softplus selective gate; random deep supervision (5 of 20 loop positions carry gradients); confidence head for early exit	3B tokens, one epoch, FineWeb-Edu	Qwen3-1.7B gains +1.6 avg points zero-shot with capabilities preserved; early exit averages 3.39 of 8 iterations at near-best accuracy	Code (GitHub thrillcrazyer/LoopUS) and one converted checkpoint (Qwen3-1.7B) public; the recipe is the point
Retrofitting Recurrence (McLeish et al., Nov 2025, arXiv 2511.07384)	Surgery on a pretrained LLM: delete middle layers, form prelude/core/coda, adapter injection, recurrence curriculum from 1 to max depth	25-52B tokens	Retrofitted TinyLlama/Llama-1B beat continued-training of the original at equal compute on math (GSM8K 52.0 vs 45.2)	Code and datasets public (GitHub mcleish7/retrofitting-recurrence; HF collection)

Family	Approach	Training budget	Key result	Availability
Ouro (Zhu et al., Oct 2025, arXiv 2510.25741; ByteDance + Bengio)	Full-stack looping: the entire 24-48 layer stack applied t times; per-step LM loss; entropy-regularized exit gate + Q-exit deployment knob	7.7T tokens (multi-stage pretraining)	Ouro-1.4B matches Qwen3-4B (GSM8K 78.9 vs 72.9; MATH500 82.4 vs 59.6); looping buys computation, not knowledge; RLVR on top failed to improve it	Apache-2.0 on Hugging Face (ByteDance/Ouro-1.4B, 2.6B, Thinking variants)

Two further 2026 entries sharpen the picture. ARLoop (3 November 2026, arXiv 2611.05234) replaces plain residual looping with attention over the loop history and shows a 1.2B looped model outperforming Huginn-3.5B across evaluated tasks, with 0.6B and 1.2B checkpoints on Hugging Face - the strongest evidence yet that looping pays off at exactly our target scale. And “Inner Loop Inference” (February 2026, arXiv 2602.14759) shows that even a frozen, untrained model can gain modestly from looping a middle block range with representation interpolation - a zero-cost fallback if full conversion training proves too expensive on Kaggle.

4.3 The skeptical evidence - what looping does not do

An honest architecture choice has to price in the negative results, and 2025-2026 produced several. Probing work on Huginn-3.5B (arXiv 2507.02199, COLM 2025 workshop) found little interpretable latent chain-of-thought: suppressing explicit CoT, recurrence depth r=32 scores 4.93 on GSM8K against 24.87 with explicit CoT - looping is not a substitute for externalized reasoning on knowledge-heavy tasks. Ouro's own controlled experiments, independently confirmed by “Beyond Parameters” (arXiv 2506.18233), show looping does not increase knowledge capacity: memorization is unchanged at fixed parameters; gains come from manipulating context. Depth extrapolation is fragile: Ouro degrades moderately beyond its trained T=4, and models trained at loop 32 keep only 22% of correct answers at loop 64 unless trained with random loop-position supervision (which restores 99.8% retention - arXiv 2610.19768). A large-scale study (“How to Loop Your Transformer,” arXiv 2610.40325, up to 1B parameters) concludes looping is not uniformly beneficial - gains saturate or reverse depending on budget allocation. And at 135M parameters (MoR, arXiv 2507.10524) the vanilla transformer simply wins. The design lesson: looping is a compute-allocation trick that helps when (a) parameters are fixed and small, (b) the task rewards iterative computation over context, and (c) training depth and inference depth are kept consistent. A trading decision model on binned numeric features is squarely case (b): there is no factual knowledge to recall, only a state to evaluate.

4.4 Why the evidence favors looping for this project

Four independent lines of 2026 evidence align with our use case. First, structured data: latent-CoT feedback tokens improved 7 of 9 time-series datasets by an average 12.63% and 23 of 27 tabular datasets (arXiv 2605.11262); looped tabular foundation models (LoopICL, arXiv 2609.36108) match far larger models with about 90% fewer parameters; ElTab (arXiv 2610.41819) matches a 12-layer TabPFN baseline with 25% fewer inference FLOPs. Second, dynamics: “Test-Time Forecasting” (arXiv 2610.19552) establishes a theoretical link between recurrent depth and in-context learning of dynamical systems, with looped one-step forecasters generalizing to out-of-distribution dynamics during autoregressive rollout - the closest published analogue to reading a price series. Third, sequential decision-making: Looped Actor (arXiv 2609.37432) shows depth-recurrent policies matching untied baselines with 16x more parameters, with loop count growing with planning difficulty. Fourth, and most telling for us: the BabyLM 2026 result “Loop Before You Leap” (arXiv 2611.06206) shows a 12-layer stack repeated four times beating a standard 48-layer transformer by 5.4 BLiMP points at matched inference compute under a strict 100M-word data budget - looping helps most when data is the binding constraint, and fifteen years of hourly gold bars is a small dataset by language-model standards. Mechanistically, an ICLR 2027 submission shows weight tying's benefit comes from gradient pooling across loop iterations - a genuine inductive bias toward reusable iterative operators.

Finally, the niche check: searches across arXiv for looped or recurrent-depth transformers applied to trading, market prediction or financial time series return nothing as of 6 November 2026. The nearest neighbors are the dynamics and tabular papers above. To our knowledge, AURUM would be the first looped-transformer trading decision model - a genuinely open research question, which is exactly why Chapter 10's validation gates and Chapter 11's mandatory baselines exist: nobody has yet demonstrated this works, including us.

5. AI Trading Systems: What the Evidence Says Works and Fails

5.1 The flagship claims do not survive scrutiny

The most-cited LLM trading framework, TradingAgents (arXiv 2412.20138, Tauric Research; actively maintained through v0.7.1 in October 2026), simulates a trading firm with analyst, bull-bear debate, trader and risk agents. Its headline result - 26.62% cumulative return (30.5% annualized) on the best of five tech mega-caps - was measured over a three-month bull-market window (January to March 2024), on five stocks, with no transaction costs in the paper. More telling is the maintenance history: throughout 2026 the maintainers fixed look-ahead and point-in-time bugs across FRED macro data, social sentiment, decision memory and Alpha Vantage feeds, and the README now states plainly that the framework is a research scaffold, “not a strategy with a fixed, replicable return,” and that two runs of the

same ticker and date can differ. The pattern generalizes. FinRL's own ecosystem papers concede the field's three killers - low signal-to-noise, survivorship bias and overfitting - and an AI4Finance paper states that DRL results which “optimistically reported increased profits in backtesting may suffer the false-positive issue due to overfitting.” Contamination-free benchmarks deliver the verdict: on StockBench (arXiv 2510.02209), which scores sequential multi-month trading, most LLMs fail to beat buy-and-hold; FINSABER (arXiv 2505.07078) shows previously reported LLM advantages deteriorate across broader cross-sections and longer horizons, with LLM strategies underperforming in bull markets and losing heavily in bears.

5.2 The evaluation protocol we adopt: Alpha Illusion, P1-P6

The single most useful document found for this project is “The Alpha Illusion” (arXiv 2605.16895, May 2026), which audits end-to-end LLM trading claims and prescribes a six-point screening protocol. Its verdict - “reported alpha from end-to-end LLM trading agents should not be treated as deployment evidence” - comes with a constructive alternative that matches our architecture: LLMs should enter as auditable information interfaces upstream of independent calibration, risk and execution modules, which satisfies the protocol by construction once final decision authority is moved out of the LLM. We adopt the protocol verbatim as our evaluation checklist.

Table 4: The Alpha Illusion P1-P6 screening protocol, adopted for AURUM

Gate	Requirement	How AURUM satisfies it
P1. Temporal integrity	No time-travel: training and features must not leak future information, including pretraining knowledge	Returns-based features only; strict embargoes; from-scratch ablation and pre-2020 control runs (Chapter 10)
P2. Dynamic universe	No survivorship or static-universe bias	Single instrument (XAUUSD) with full bear regimes in test; no delisting issue
P3. Counterfactual robustness	Performance must not flip under reversed evidence; no hidden style tilt	Reverse-evidence direction-flip tests; per-regime reporting; regime-conditional calibration
P4. Epistemic calibration	Language confidence is not tradable probability; verify ECE and reliability per regime	Calibration loss in training; monthly temperature-scaling refits; reliability monitoring per regime
P5. Realistic implementation	Gross-alpha illusion: token, latency, spread, swap and slippage costs must be netted	Full cost model in backtest (Chapter 8); per-decision cost ~0.001 cents at 0.6B scale
P6. Multi-agent disaggregation	Consensus among correlated agents is not independent confirmation	Single model, single decision policy; no echo chambers; deterministic execution

5.3 Foundation models for time series are not a free lunch

Zero-shot time-series foundation models - the obvious alternative backbone - have documented blind spots. Chronos-2 is the most downloaded time-series model on Hugging Face (27.1M monthly downloads), yet SimpleTimeBench (arXiv 2610.02058, October 2026) shows Chronos-2, Moirai 2.0 and Toto frequently produce suboptimal forecasts even for trivial monotonic trends, periodic signals and leading-indicator covariates. A 30-dataset deployment study (arXiv 2607.04919) finds foundation models beat classical methods (naive, ETS, ARIMA, XGBoost) on only 15 of 30 datasets, with classical methods winning on some datasets using 2% of the data. And ablation work on LLM-based forecasters (arXiv 2406.16964) shows removing the language model or replacing it with a basic attention layer does not degrade performance. The tabular evidence is equally blunt: tree ensembles still beat deep models on medium-sized numeric tabular data (Grinsztajn et al., arXiv 2207.08815; McElfresh et al., arXiv 2305.02997). The consequence is a hard requirement, not a preference: AURUM's looped transformer must beat LightGBM/XGBoost and a small GRU trained on identical features under identical cross-validation, or it does not ship. The model earns its place only if the looped architecture's iterative computation adds measurable decision value over cheaper alternatives.

5.4 The one differentiator that keeps appearing: online learning

Across verified competition winners (detailed in Chapter 7), the recurring “secret sauce” is not architecture but adaptation: updating model weights after deployment, at low learning rates, on newly arrived data. The second-place solution of the 2024-25 Jane Street competition derived nearly a third of its final score from intraday online gradient updates on top of a frozen 17-model ensemble; the solo winner of Optiver's 2023-24 competition applied online learning to every component of a CatBoost-plus-GRU-plus-Transformer blend. A static model decays; markets drift. AURUM's plan therefore includes a weekly online-learning stage on Kaggle with strict purging of overlap between the update window and the evaluation window. This is also where the looped architecture may pay a second dividend: loop-count and iteration-divergence signals give unusually cheap drift detectors for deciding when recalibration or retraining is needed.

6. The Best Results Ever Achieved by Individuals

The user's first research priority was the best ever verified from individual builders. The verified record, across four independent arenas, is remarkably consistent - and consistently modest.

6.1 Kaggle finance competitions

The Jane Street Real-Time Market Data Forecasting competition (October 2024 to July 2025; 25,703 registrations, 2,781 teams) is the largest recent test of individual skill on financial prediction. The private leaderboard's top ten is dominated by single-person teams (ms capital, first at 0.013890 weighted R-squared; Patrick Yam, second; hyd, fifth). Note the magnitude: the winning edge is an R-squared of 1.4% - the best individuals on earth, on a clean feature set, extract a sliver of signal. The verified second-place method is a 17-model neural ensemble with online learning: a reproduction by a third party measured the frozen ensemble at 0.01413 and the online-updated version at 0.01942, i.e., +0.005 absolute - a third of total performance - from post-deployment adaptation. The verified eighth-place solution used GRU sequence models and reported explicitly that “MLP, time-series transformers, cross-symbol attention and embeddings didn't work for me.” In Optiver's Trading at the Close (2023-24), the solo winner “hyd” - a repeat winner, also fifth at Jane Street - blended CatBoost (0.5), GRU (0.3) and Transformer (0.2) with online learning and post-processing. The historical Jane Street 2021 winner was a team, but its core was a supervised autoencoder-plus-MLP trained with a 31-gap purged 5-fold time-series split - validation hygiene, not exotic architecture, was the differentiator.

6.2 Numerai, and the honest professionals

Numerai's staking leaderboard is the largest set of verified, on-platform, real-money track records for individuals in quant. As of 6 November 2026: 568,149 NMR staked across 679 staked participants, with an average one-year return of +19.97%. The top of the all-time board: bpa at 52.1% one-year on 6,185 NMR staked; derrida at 47.2% on 6,417 NMR. These are real and verified - but the edge belongs to Numerai's metamodel; individual skill expresses itself as model selection and staking strategy, and self-selection means the average hides the dead. The cleanest professional benchmark remains Rob Carver, the ex-AHL manager who publishes his own live systematic results: 2014-2023 at a mean +14.1% per year on 24.3% annualized volatility - a live Sharpe ratio of 0.58 - against the SG CTA index at 0.38 over the same window, including a -8.7% year in 2022-23. That is what a world-class retail systematic trader with 114 instruments, a 50-year backtest and decades of experience actually achieves.

6.3 The failure statistics

The base rates are harsh and well documented. Regulator data: the FCA's own analysis found 82% of retail CFD clients lose money (average loss 2,200 pounds per client per year); France's AMF reports 89% losing (average 10,887 euros); OANDA's own website footer currently discloses 76.6%. Day-trading studies: fewer than 1% of Taiwanese day traders are persistently profitable net of fees (Barber, Lee, Liu and Odean, Journal of Financial Markets 2014), and 97% of Brazilian futures day traders who persisted past 300 days lost money

(Chague, De-Losso and Giovannetti, SSRN 3423101). One 2025 paper documents the exact failure mode this project must avoid: a 500-agent crypto trading system that validated at 300%+ APY and then lost more than 70% of live capital (“The Red Queen's Trap,” arXiv 2512.15732).

Table 5: Verified individual performance ceilings (net of costs)

Arena	Best verified individual result	Sharpe / edge	Confidence
Jane Street 2024-25 (Kaggle)	Solo and near-solo teams fill the top 10; win = ensemble + online learning	R-squared 0.014	HIGH (leaderboard + writeups)
Optiver TAC 2023-24 (Kaggle)	Solo winner “hyd”: CatBoost+GRU+Transformer blend + online updates	RMSPE 5.40 private	HIGH (writeup)
Numerai staking	bpa: +52.1% one-year on 6,185 NMR staked; 679 staked average +19.97%	n/a (payouts)	HIGH (on-platform, live)
Systematic futures (blog, live)	Rob Carver 2014-2023: +14.1%/yr at 24.3% vol incl. -8.7% year	SR 0.58 live	HIGH (fetched)
Day trading (studies)	Under 1% persistently profitable (Taiwan); 97% of persistent Brazilian day traders lose	n/a	HIGH (published)
XAUUSD-only systematic track record	None found, by any individual or lab	n/a	HIGH (absence documented)

The last row is the strategic summary: nobody, anywhere, publishes a verified live track record for a systematic XAUUSD-only strategy. There is no proof it cannot be done - only proof that if it is done, it will be done for the first time. The engineering conclusion is to set the success bar at a net-of-cost Sharpe of 0.5 to 1.0 (Carver-class), plan for 20%+ drawdowns, treat anything above SR 1.0 in backtest as a multiple-testing artifact until the Deflated Sharpe Ratio says otherwise (Chapter 10), and let the system prove itself in months of paper trading before a single dollar is risked.

7. Market Reality: Gold in November 2026

7.1 Where the market stands

Gold closed this research day around $4,053-4,058 (spot $4,053.50 at 15:22 UTC, 6 November 2026; COMEX December futures $4,054). That number sits 27.4% below the all-time high of $5,586.20 printed on 29 January 2026, after a violent correction to $3,879.30 on 30 October - a max drawdown of 26.7% peak-to-trough that is still resolving. Realized volatility is 29.5% annualized in 2026, against 20.9% in 2025 and 15.3% in 2024. The average daily range is $80-120 (2.0-2.6% of price), and 2026 year-to-date stands at -6.5%. Demand-side, the World Gold Council's Q3 2026 report shows total demand of 1,369 tonnes

(-1% year-on-year) but a record value of $188 billion, with central banks buying 321 tonnes in the quarter (+42%, the fastest since late 2024, 675 tonnes year-to-date) and ETF holdings at a record 4,256 tonnes after +211 tonnes of quarterly inflows. In short: the training and test data for this project contain a once-in-decades parabolic rally and a live crash - the regime stress test most trading research never gets.

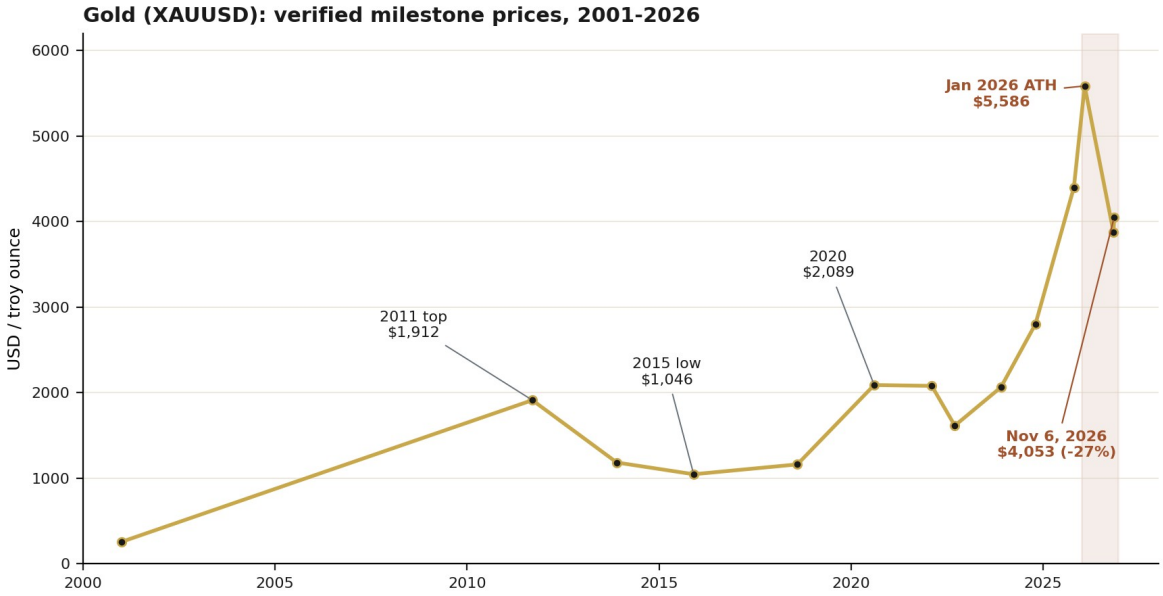


Figure 1: Verified gold price milestones, 2001-2026. Milestone-to-milestone path through verified COMEX/spot data points (research thread E); shaded band marks the 2026 drawdown regime in which the project begins.

7.2 The full regime map

Table 6: Gold regimes 2001-2026 (verified price data, research threads D and E)

Period	Price path	Regime character and driver
2001-2011	$255 to $1,912 (Sep 2011)	Secular bull: USD bear, QE, collapsing real yields, crisis hedging
2013	$1,695 to $1,183 (-28%)	Taper tantrum; ~500t ETF liquidation
2014-2018	$1,046 (Dec 2015) to $1,375 range	Low-vol, range-bound grind; rising real yields, strong USD
2018-2020	$1,161 to $2,089 (Aug 2020)	Fed pivot plus COVID: real yields below -1%, QE, ETF inflows
2021-2022	$1,614-2,079 chop	Hiking cycle; Ukraine-invasion spike $2,079 (Mar 2022), USD spike low $1,614
2023	$1,810-2,063 range	Range with geopolitical bid (Oct 2023)
2024	Breakout: $2,234 (Mar) to $2,802 (Oct)	Regime change begins: central-bank buying above 1,000t/yr, de-dollarization
2025	$2,838 (Jan) to $4,398 (Oct)	Accelerating mega-rally; vol steps up 15% to 21%
2026	ATH $5,586 (Jan 29), crash to $3,879 (Oct 30), now ~$4,053	Parabolic blow-off then -27% correction; realized vol 29.5%; record CB and ETF holdings

The machine-learning consequence is symmetric and brutal: a model trained on 2013-2018 learns a mean-reverting, range-bound asset and would be destroyed by 2024-2026; a model trained through 2025 learns that volatility ratchets and price only rises, and would be destroyed by the -27% drawdown that followed. Both failure modes are inside our data. Chapter 10 therefore mandates regime-blocked training splits and per-regime reporting: a strategy that only works in trending bull regimes is leveraged beta, not alpha.

7.3 Costs, sessions and events - the numbers the backtest must charge

Table 7: XAUUSD microstructure facts (2025-2026, research thread D)

Item	Value	Implication
All-in round-turn cost, raw/ECN account	$0.15-0.35 per oz (spread $0.05-0.25 plus ~$6-7/lot commission)	About 1-3 basis points of price; negligible for multi-hour holds, fatal for scalping
All-in cost, standard account	$0.30-0.60 per oz (spread only)	Use $0.30-0.35 as the base case in backtests; stress at $0.60
Swap (rollover), long positions	About -$0.35/oz per night (-$30 to -$45 per lot); triple on Wednesday	A 3-day long hold costs ~$1/oz (~2.5bp) - must be modeled as explicit overnight cashflow
Average daily range (2026)	$80-120 (2.0-2.6%); ATR14 ~ $85	Even 0.25x-ADR targets are ~100x the round-turn cost
Session ranges (avg $ per hour, UTC)	Asia $14.8, London $15.4, NY $17.7; peak 13:00-14:00 at $24.3 (0.67%)	Peak = US data releases plus COMEX open; event-time features beat session dummies
Weekend gaps (10y, 451 Mondays)	Mean 0.32%, median 0.19%, p95 1.09%, max 3.52%	Stops do not guarantee fills; Friday-risk rules and gap-aware risk budgeting required
Event calendar	FOMC statements 19:00 UTC (8/yr); NFP and CPI 13:30 UTC	Free calendars: federalreserve.gov (verified), BLS schedule pages, FRED release API
Retail leverage caps (EU/UK)	Gold 20:1 (5% margin); 0.01-lot minimum margin ~$202 at $4,000 gold	Minimum viable live account ~$500-2,000; ~$7,500 for strict 1% risk with $75 stops

8. Component Selection: Base Model, Data and Backtesting Stack

8.1 Base model

Table 8: Base model candidates at or under ~2B parameters (November 2026; parameter counts and licenses from Hugging Face configs, tokenizer behavior tested empirically)

Model	Params	License	Context	Notes
Qwen3-0.6B(-Base) - recommended	0.75B	Apache-2.0	32k	21.5M downloads/mo; digit-by-digit tokenizer (verified); LoopUS-proven family; best throughput per GPU-hour
Qwen3.5-0.8B(-Base) - alternate	0.87B	Apache-2.0	262k	March 2026 release; AA Intelligence Index 9 vs 6.5 for Qwen3-0.6B (MEDIUM); 2-KV-head GQA gives tiny KV cache
Qwen3-1.7B(-Base)	2.03B	Apache-2.0	32k	The exact backbone LoopUS converted; 2.5-3x slower than 0.6B on T4 - reserve as fallback
Llama-3.2-1B	1.24B	Llama 3.2 (gated)	131k	3-digit-group tokenizer (verified) - worse for numeric features; license friction
SmolLM2-1.7B	1.71B	Apache-2.0	8k	Digit tokenizer but weaker benchmarks and short context
OLMo-2-1B	1.49B	Apache-2.0	4k	Fully open data; 3-digit groups; 4k context too short
Gemma-3-1b-it	1.00B	Gemma terms	32k	Restrictive terms; fine tokenizer; instruction-tuned only
Pythia-1.4b	1.51B	Apache-2.0	2k	2023-era quality; arbitrary digit merges - excluded

Recommendation: Qwen3-0.6B-Base as the primary backbone, with Qwen3.5-0.8B-Base as the alternate if its Unsloth support matures. The decisive factors are empirical: Qwen tokenizers split numbers digit-by-digit (verified by direct tokenizer execution), which is the encoding LLMTime (arXiv 2310.07820) found materially better for numeric work than grouped-digit schemes; the Qwen3 family is the backbone LoopUS actually converted; and at 0.75B parameters it maximizes looped-training tokens per Kaggle GPU-hour (Chapter 9). Numeric features will enter as Chronos-style binned tokens (arXiv 2403.07815) - each feature scaled and quantized to a small integer vocabulary - keeping sequences short and robust; exact scaled-digit encoding remains an ablation option.

8.2 Data

Table 9: Free data sources for XAUUSD and correlated series (verified 6 November 2026)

Source	XAUUSD coverage	Granularity	Role in the stack
Dukascopy (via dukascopy-node v1.51.1)	From Jun 1999 (per tool table) / 2003 (official) - present	Tick, M1-H4, D1	PRIMARY: ~600-800 MB of M1 for 20 years, uploaded as a private Kaggle dataset
HistData.com	March 2009 - present	M1 + tick (bid)	Cross-check venue; note EST timezone in ASCII files

Source	XAUUSD coverage	Granularity	Role in the stack
Yahoo Finance (yfinance 1.8.0)	GC=F futures only - XAUUSD=X is now DELISTED; 1h limited to last 730 days, 15m to 60 days, 1m to 8 days/request	1m-1mo	Recent-intraday features + cross-check; EURUSD=X, SI=F, DX-Y.NYB verified working at 1h
FRED (free)	GOLDPMGBD228NLBM (LBMA PM fix, 1968-); DGS10, VIXCLS, DCOILWTICO, DEXUSEU, DTWEXBGS, SP500	Daily	Macro and correlated daily features (US10Y, VIX, WTI, EURUSD, DXY, SPX)
OANDA v20 practice (free)	XAU_USD; 5,000 candles per request, paginated	S5 to M	Forward/paper trading environment and live feature parity
Federal Reserve / BLS calendars	FOMC dates (verified server-rendered HTML through 2027); NFP/CPI at 13:30 UTC	Events	Event filters and post-event volatility features

Two data warnings carry engineering weight. First, Yahoo's spot symbol XAUUSD=X is delisted - any notebook or dataset built on it is silently broken - and COMEX futures (GC=F) are not back-adjusted, so their quarterly roll gaps contaminate long-horizon studies; futures data is used only for the recent-730-day feature panel, never for the primary series. Second, Dukascopy pre-2003 XAUUSD tick density is uncertain, so the primary panel starts conservatively at 2003 with a bar-density audit before any earlier data is admitted.

8.3 Backtesting stack

The research loop uses vectorbt 1.2.0 (released 19 October 2026; actively maintained): vectorized execution over ~100k H1 bars in seconds, parameter sweeps, and clean composition with quantstats 0.0.86 and the statistics tooling below. NautilusTrader 1.231.0 provides the final realism pass - true bid/ask fills from Dukascopy bid and ask series, latency and partial-fill models - before any paper-trading claim is made. Swap is modeled as an explicit overnight cashflow (-$0.35/oz/night long, triple Wednesday) and walk-forward as a simple outer loop. backtesting.py 0.6.6 is a acceptable minimal fallback; zipline-reloaded (equity-daily paradigm) and freqtrade (crypto venues) are wrong tools for gold. For the statistical layer: mlfinlab is dead (commercial since 2023), replaced by purgedcv (PyPI, updated 4 November 2026) for purged/embargoed and combinatorial CV, and ml4t-diagnostic for Deflated Sharpe Ratio, probability of backtest overfitting and White's Reality Check. One caution from the deflated-sharpe package's own case study: a grid search of 19,200 configurations on BTC 1-hour walk-forward produced many in-sample Sharpe 1.5+ strategies and the DSR rejected every one; the binding constraint was trade count, not search method.

9. Free-Stack Training Feasibility on Kaggle

9.1 What the free tier actually provides (November 2026)

Kaggle retired the P100 on 15 September 2026: the free GPU offering is now a single machine shape, 2 x Tesla T4 (15.4 GB each), with 4 CPU cores and 29 GB RAM. The weekly quota is 30 GPU-hours from a shared pool - one hour on both T4s counts as one hour - resetting Saturdays at 00:00 UTC, with sessions capped at 12 hours, a 20-minute interactive idle timeout, 20 GB of persistent output in /kaggle/working, and internet access (phone-verified accounts). A separate TPU v3-8 pool offers roughly 20 additional hours per week; adapting the looped architecture to torch/xla is unproven and treated as upside, not plan. Two T4s with Accelerate DDP deliver a measured 1.8-2.4x speedup on sub-2B fine-tunes in public benchmarks. FlashAttention-2 does not run on Turing GPUs; PyTorch SDPA (used automatically by Unsloth, which officially supports T4's CUDA capability 7.0) is the attention backend.

9.2 Throughput math, and the one big adjustment

Published Unsloth benchmarks on a free-tier T4 show a 1.1B model at 4-bit QLoRA training 3.87x faster than plain Hugging Face (-73.8% VRAM); scaling to our sizes gives working estimates of 3,000-6,000 tokens/second for 0.6B and 1,500-2,500 for 1.7B, roughly doubling with both T4s. A 30-hour GPU-week therefore buys roughly 600M-1.2B tokens of standard QLoRA training on the 0.6B backbone. Looped training multiplies per-token cost: LoopUS at B=20 loop steps with K=5 supervised depths costs an estimated 4-8x per token (derived from the paper's mechanics - the single biggest uncertainty in this chapter). Figure 2 shows the resulting feasibility picture: the paper's exact recipe (3B tokens on a 1.7B backbone) would take 23-81 weeks of free quota and is out of reach; a scaled recipe on the 0.6B backbone at B=8, K=3 fits 160-475M tokens per week.

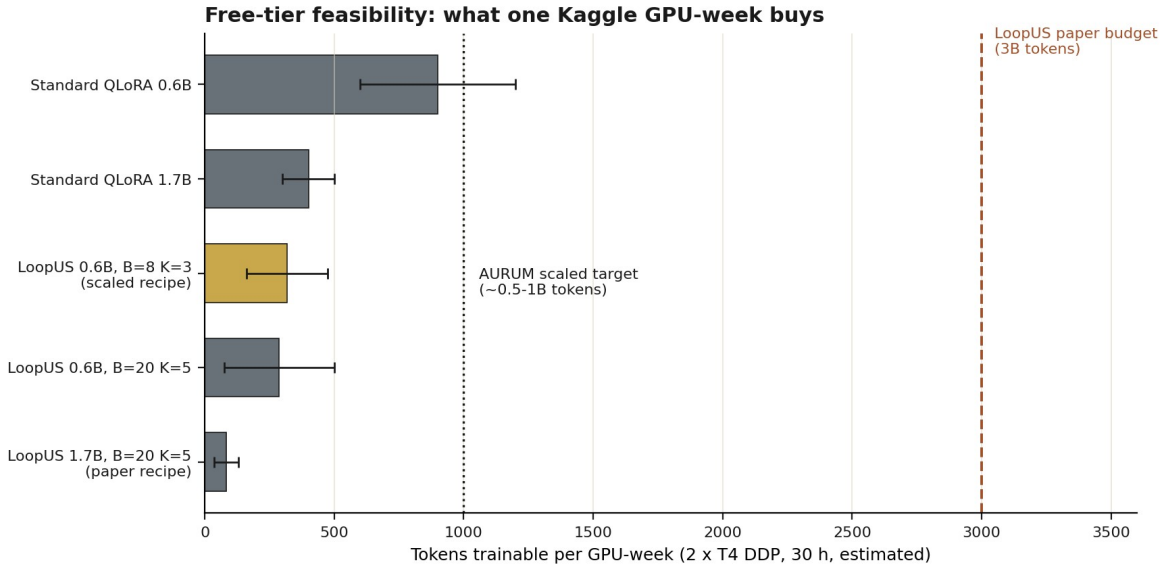


Figure 2: Tokens trainable per Kaggle GPU-week by scenario (midpoints with low-high ranges; Unsloth-relative throughput estimates, research thread C). LoopUS's 3B-token language-retention budget is out of free-tier reach, but AURUM's ~0.5-1B token budget is not.

The crucial adjustment: LoopUS needed 3B tokens because its goal was to add looping while retaining general language capability on web text. AURUM has no such obligation - the model only ever needs to read binned market-state features and score fixed actions. The conversion can therefore be trained directly on task data: hundreds of millions of tokens of market states (each hour of 2003-2026 history yields many overlapping training examples) plus a small generic replay to stabilize the loop. At 0.5-1B tokens on the 0.6B backbone, the scaled conversion fits inside 2-4 GPU-weeks with margin - provided a 30-minute throughput smoke test on Kaggle, run as Phase 0 of the build plan, confirms the estimated looped tokens/second before any multi-week commitment.

Table 10: Kaggle free-tier facts and their engineering consequences

Constraint	Value (Nov 2026)	Engineering consequence
GPU shape	2 x T4 16GB (P100 retired Sep 15, 2026)	All VRAM math done for 15.4 GB; no FA2 - SDPA attention; fp16 (no bf16) on T4
Weekly quota	30 GPU-hours, shared pool, resets Sat 00:00 UTC	Checkpoint every N steps to /kaggle/working and design all runs resumable - a crash must never cost a week
Session cap	12 hours; 20-min interactive idle timeout	Long jobs via CLI push / Save & Run All in ~10.5h segments with resume
RAM / CPU	29 GB / 4 cores	Pre-tokenize datasets once; avoid CPU-bound data loading with 2 DDP workers
Persistent output	20 GB /kaggle/working	LoRA adapters (~100-500 MB) and final 0.6B fp16 checkpoint (~1.2 GB) fit comfortably; keep 2-3 checkpoints max
DDP speedup	1.8-2.4x measured on sub-2B models	Training script as .py launched via accelerate, not in-notebook (CUDA fork issue)

Constraint	Value (Nov 2026)	Engineering consequence
Inference latency	0.6B looped (B=8) on T4: ~2-15 s per decision; CPU int4: 7-40 s (borderline)	Hourly decision cadence needs under 60 s - T4 comfortably, CPU only as fallback

10. Validation Protocol and Risk Management Framework

10.1 The statistics that kill backtests

Every quantitative claim in this section follows Bailey and Lopez de Prado's framework (The Deflated Sharpe Ratio, Journal of Portfolio Management 2014; The Probability of Backtest Overfitting, Journal of Computational Finance 2016). Three numbers anchor the design. First, minimum track length: for a strategy trading 250 times per year, an annualized Sharpe of 0.5 needs about 2,707 trades (roughly 10.8 years) before the observed Sharpe is statistically greater than zero at 95% confidence; even an excellent SR 1.0 needs 678 trades (~2.7 years). Second, the multiple-testing trap, shown in Figure 3: after trying 100 configurations on 1,500 trades, pure noise is expected to produce a best backtest Sharpe near 1.0. The 19,200-configuration case study in Chapter 8 is this trap in the wild - every Sharpe 1.5+ in-sample result was rejected by the Deflated Sharpe Ratio. Third, effective sample size: fifteen years of H1 bars is ~87,000 bars, but a 24-hour-horizon label leaves only ~3,600 effectively independent decision points, and the binding unit is trades, not bars. The protocol consequences are non-negotiable: combinatorial purged cross-validation with embargo (purgedcv) instead of a single walk-forward path; every trial counted - features, horizons, hyperparameters, losses, prompts included; Deflated Sharpe Ratio at or above 0.95 and probability of backtest overfitting below 0.5 as release gates; and pre-registration of the final configuration before the test window is touched.

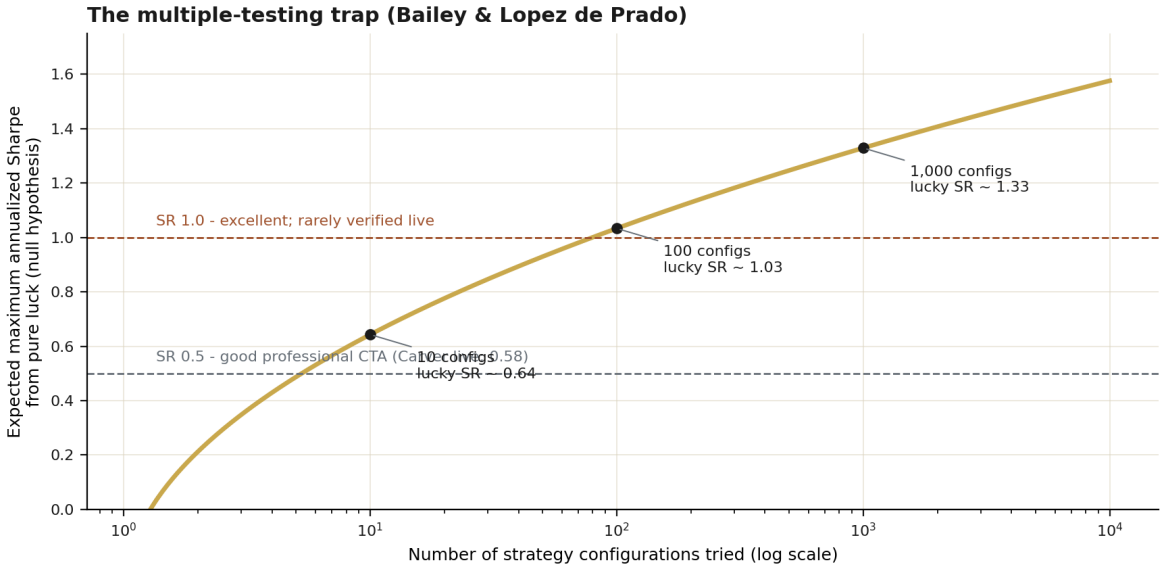


Figure 3: The multiple-testing trap. Expected maximum annualized Sharpe produced by pure luck as a function of configurations tried (1,500 trades assumed). After ~100 configurations, noise alone yields a backtest Sharpe near 1.0 - the reason every trial must be counted and the Deflated Sharpe Ratio applied.

Table 11: Minimum trades required for statistical significance (95%, one-sided)

Annualized Sharpe	Min. trades (MinTRL)	Years at 250 trades/yr
0.5 (good professional level)	2,707	~10.8
1.0 (excellent; rarely verified live)	678	~2.7
1.5 (rare; usually overfit)	302	~1.2

10.2 Data splits, contamination controls and calibration

Data is split by regime blocks, never randomly: training on 2003-2018 (two full bull-bear cycles), validation on 2019-2022, and a locked test on 2023-2026 which contains both the mega-rally and the -27% crash, with embargoes at every boundary. Contamination from the pretrained backbone is treated as a first-class experimental variable, because a 2024-2025-pretrained model has literally read about gold's rally: all features are returns-based or binned (never raw price levels, which would date-stamp the observation); a from-scratch, randomly-initialized control model of identical architecture runs on identical data - if the pretrained init beats it specifically in the 2020-2026 window, that is leakage evidence; a pre-2020-only training run serves as the leakage-free control; and a zero-shot probe quantifies what the base model already knows about gold price history. Calibration is monitored, not assumed - RLHF demonstrably destroyed GPT-4's calibration (ECE 0.007 pre-training vs 0.074 post-RLHF, from the GPT-4 technical report), and the 17-day Jev experiment measured confidence running backwards in the field. The deployed protocol: expected calibration error and reliability diagrams per regime bucket, monthly temperature-scaling refits on a rolling

window, and a pre-registered confidence-ablation test - confidence-weighted sizing must beat flat sizing on the same signals, or the confidence channel is noise and is switched off. Sizing follows the meta-labeling pattern (Lopez de Prado, Advances in Financial Machine Learning): the model proposes direction and probability; a secondary calibrated probability sizes the trade; miscalibrated probabilities are miscalibrated leverage.

10.3 Risk management: sized for the edge we can prove

Table 12: Risk framework for AURUM (rules of the deterministic execution layer)

Rule	Setting	Basis
Risk per trade	0.5-1.0% of equity (hard cap 1%)	Fixed-fractional standard; below quarter-Kelly for p=0.55-class edges (full Kelly would be 10%)
Position granularity check	0.01 lot (1 oz); at $4,000 gold a $75 stop risks $75 = 1.5% of a $5,000 account	Minimum position can exceed the risk budget on small accounts - account size must fit stop width
Volatility targeting	Scale gross exposure to ~15% annualized target	Gold realized 29.5% in 2026 vs 15.3% in 2024 - fixed size would have tripled risk
Stops	ATR-based (1.5-3x ATR14, ~$128-255 at current vol)	Adapts to the 2x vol regime shift; fixed-dollar stops fail across regimes
Drawdown circuit breaker	At -15% from equity high: flat, recalibrate on paper, human review to re-arm	The real killer of retail systems is abandonment at -20%; pre-commit the shutdown rule
Event and gap rules	No new entries in the 30 minutes around FOMC/NFP/CPI (13:30/19:00 UTC); reduce or flat into event weekends; Friday risk budget halved	Weekend gap p95 1.09%, max 3.52%; stops do not guarantee fills across gaps
Leverage and account	EU/UK 20:1 on gold; micro-lot live minimum ~$202 margin per 0.01 lot; realistic live account $2,000-7,500	FCA PS19/18 margin floors; keeps 1%-risk budget consistent with stop width
Go-live sequence	CPCV backtest, then DSR/PBO gates, then 3-6 months and 100+ trades on OANDA practice with live calibration monitoring, then micro-lots at 0.25% risk; scale only after 300+ live trades with Sharpe CI excluding zero	Every documented failure in Chapter 6 skipped at least one of these steps

11. Master Limitations Table and Mitigations

The user asked explicitly for the limitations of the plan, found before building. Twelve survive the research, each paired with its mitigation and the residual risk that remains after the mitigation. Items 5, 9 and 12 are the ones most likely to end the project in its original form, and they are the reason the build plan gates every phase.

Table 13: Limitations, mitigations and residual risk

#	Limitation (with evidence)	Mitigation (built into the plan)	Residual risk
1	Backtest overfitting: ~100 tried configs produce a lucky SR ~1.0 at achievable sample sizes (Ch. 10)	CPCV + embargo; count every trial; DSR >= 0.95 and PBO < 0.5 release gates; pre-registration	A single-instrument strategy may never gather 700+ clean post-selection trades - the honest edge may be unprovable
2	Regime shift: 2024-26 rally and -27% crash both in-window; models trained on any sub-period learn the wrong lesson (Ch. 7)	Regime-blocked splits; per-regime reporting; vol targeting; weekly online learning (the verified winners' differentiator)	The next regime is, by definition, not in the data
3	Pretraining contamination: the base model has read about 2024-26 gold (Ch. 10)	Returns-based binned features only; from-scratch control; pre-2020 control run; zero-shot knowledge probe	Leakage through subtle priors cannot be fully excluded
4	Calibration inversion: Jev's field confidence correlated -0.17 with outcomes; RLHF wrecked GPT-4 calibration (Ch. 3, 10)	Calibration-penalized training objective; per-regime ECE monitoring; monthly refits; confidence-ablation kill switch	Calibration under live regime shift may still decay between refits
5	GBDTs may simply win: trees beat deep models on medium numeric tabular data; LLM backbones added nothing in TS ablations (Ch. 5)	Mandatory LightGBM/XGBoost and GRU baselines under identical CPCV; the looped model ships only if it beats them	High: the honest outcome may be that the looped transformer loses - then the baseline system ships instead
6	Kaggle quota fragility: 30 GPU-h/week, 12h sessions, T4-only since Sep 2026 (Ch. 9)	Resumable runs, checkpointing, CLI-pushed batch segments, pre-tokenized data; TPU pool as unproven upside; Colab fallback	A multi-week conversion run disrupted mid-way costs calendar weeks, not just GPU-hours
7	Free data quality: roll gaps in futures, delisted Yahoo spot symbol, thin pre-2003 Dukascopy ticks, EST-timezone HistData files (Ch. 8)	Dukascopy M1 primary from 2003; HistData cross-venue audit; bar-density checks; UTC normalization	Silent gaps or bad ticks in any year could bias features; audits catch most, not all
8	Execution realism: spread widens at events; weekend gaps (p95 1.09%, max 3.52%); swap drag ~2.5bp for 3-day holds (Ch. 7)	Full cost model in vectorbt; NautilusTrader bid/ask realism pass; event and Friday rules; swap as explicit cashflow	Live slippage during vol spikes can exceed any modeled spread
9	Small effective sample: ~3,600 independent H1 decision points and 50-200 trades/yr at swing cadence (Ch. 10)	20+ years of M1 data; multi-horizon experiment matrix (M15/H1/H4/D1) to find the best trade-count/edge balance	High: at 100 trades/yr, a true SR 0.5 needs a decade to verify - paper trading is the only honest intermediate evidence
10	Single instrument: no diversification; gold-only systems have no verified public precedent (Ch. 6)	Phase 1 is deliberately single-instrument for research cleanliness; architecture keeps a portfolio dimension for later FX/metal expansion	Idiosyncratic gold regime risk is fully concentrated
11	Loop-specific fragilities: depth extrapolation degrades; retention collapse at unseen loop counts; naive quantization breaks looped models (Ch. 4)	Train and serve at consistent depths; random loop-position supervision (99.8% retention in the source study); loop-aware quantization or none	Looped-model serving is a young systems literature; T4 deployment may need engineering patience

#	Limitation (with evidence)	Mitigation (built into the plan)	Residual risk
12	No precedent, no proof: no looped-transformer trading model exists; nobody publishes a verified XAUUSD-only track record (Ch. 4, 6)	Treat the project as research with a strict publication-grade evaluation protocol; baselines ship value even if the looped model loses; no live capital before the full gauntlet	The idea itself may not work - which is exactly what the gates are for

12. Recommended Architecture and Phased Build Plan

12.1 The AURUM-0.6B system

AURUM (Latin for gold) is a System One decision model in the Jev mold, built open and self-hosted. Data plane: Dukascopy XAUUSD M1 bid/ask from 2003, resampled to H1 and D1; features are returns-based, volatility-normalized and Chronos-binned into the Qwen tokenizer's vocabulary - trend and momentum percentiles, ATR-relative stretch, session and event flags, funding-style proxies from futures basis, and correlated daily series (DXY, US10Y, VIX, WTI, EURUSD, silver, SPX from FRED and Yahoo). No raw price levels ever enter the state, and no free text enters at all, which closes Jev's prompt-injection failure mode by construction.

Model: Qwen3-0.6B-Base receives LoopUS-style surgery - encoder = first layers, a middle-block loop group behind a softplus selective gate, decoder = final layers - configured at B=8 loop steps with K=3 randomly supervised depths for memory-efficient training, plus a confidence head for early exit and the KL-between-successive-loops divergence as a free uncertainty signal. Heads are parallel and typed, not autoregressive: a direction choice (long / short / stand-aside), a conviction noul, a target-distance score, and a regime-noul; option symmetry is enforced architecturally (a single scalar per direction rather than ordered options) so the option-order bias measured at 26% in the Jev experiment cannot arise. Training runs in three stages: Stage A converts the backbone into a stable looped model on mixed data (~0.5-1B tokens: market states plus a small generic replay); Stage B trains the decision heads with an RLCD-style objective - proper scoring rules (Brier/log-loss) plus an explicit calibration-penalty term, class-balanced across regimes; Stage C is weekly online learning at low learning rate on the newest data with strict purging - the one differentiator that every verified individual winner shared. All training on Kaggle 2 x T4 via Accelerate DDP with resumable checkpoints; the model judges, and the deterministic execution layer of Table 12 decides and executes.

The evaluation gauntlet is the point of the whole system: mandatory baselines (buy-and-hold, time-series momentum, LightGBM, a small GRU) under identical combinatorial purged CV; the P1-P6 checklist; DSR at or above 0.95 and PBO below 0.5; full-cost vectorbt backtest and a NautilusTrader realism pass; then an OANDA practice phase with live calibration

monitoring. The model ships only if it beats every baseline net of costs under the same splits - and if it does not, the baseline system ships and the looped model remains the research track.

12.2 The seven phases

Table 14: Build plan - phases, deliverables and GO/NO-GO gates (Kaggle GPU-weeks are estimates from Chapter 9)

Phase	Weeks	Deliverables	GO/NO-GO gate
P0. Smoke tests	1 (0.5 GPU-wk)	Kaggle T4x2 throughput benchmark of the looped forward/backward pass; Dukascopy download test; resume-from-checkpoint test	Measured looped throughput confirms >= 150M tokens/week; resume works after kill
P1. Data infrastructure	1-2	Dukascopy M1 2003-2026 (~700 MB) via dukascopy-node; UTC-normalized H1/D1 panel; feature pipeline; HistData cross-audit; private Kaggle dataset; event calendar table	Bar-density audit passes; cross-venue divergence quantified; dataset versioned on Kaggle
P2. Baseline gauntlet	2-3	CPCV + DSR/PBO tooling (purgedcv, ml4t-diagnostic); buy-and-hold, TSM, LightGBM, GRU baselines on identical features	Validation harness reproduces baseline scores; the bar to beat is registered and frozen
P3. Looped conversion	3-6 (2-4 GPU-wk)	LoopUS surgery on Qwen3-0.6B-Base; Stage A conversion training with checkpointing; convergence and loop-stability diagnostics	Looped model matches or beats the GRU baseline on validation proxy tasks at B=8; no divergence at trained depth
P4. Decision training + calibration	6-8 (1-2 GPU-wk)	Stage B decision-head training; per-regime reliability diagrams; temperature scaling; confidence-ablation A/B	ECE < 0.05 in every major regime bucket; calibrated probability beats LightGBM on the decision metric
P5. Validation gauntlet	8-10	Full CPCV + DSR + PBO; full-cost vectorbt backtest; NautilusTrader realism pass; contamination ablations (from-scratch and pre-2020 controls)	DSR >= 0.95; PBO < 0.5; >= 700 post-selection trades; net-of-cost Sharpe beats all baselines; P1-P6 checklist green
P6. Paper trading	10-16+	OANDA practice deployment (hourly cadence); weekly online updates; live calibration and drift monitoring; trade log	>= 100 paper trades; net paper Sharpe > 0; ECE stable; only then discuss micro-lot live at 0.25% risk

12.3 What success looks like - honestly

Success is not a money printer; it is a system whose edge is either proven or refuted with publication-grade statistics, entirely on free infrastructure. The base case after 16 weeks: a validated backtest meeting the DSR/PBO gates at a net-of-cost Sharpe between 0.3 and 0.8, in paper trading, with calibration holding per regime. The stretch case: the looped transformer beats LightGBM decisively and the loop-depth signal demonstrably encodes regime difficulty - a first-of-kind result worth publishing either way. The failure case, which the gates make cheap: the looped model loses to gradient boosting, and the project still

delivers a rigorously validated gold strategy stack plus a reusable looped-transformer training pipeline on Kaggle. Capital is not part of the success criteria; Chapter 6's evidence says that is the only honest ordering.

13. Key Sources and Confidence Notes

Sources are grouped by load; arXiv identifiers are given where a paper is the origin of a claim. Confidence follows the convention of Chapter 2: HIGH marks primary sources read in full during this research; MEDIUM marks secondary or multi-source agreement; LOW marks single unverified claims, all of which are listed in 13.5. The DeerFlow methodology reference is the repository cloned at the start of this research (github.com/bytedance/deer-flow, version 2.0, MIT license).

13.1 Architecture (looped transformers)

- Geiping et al., Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach, arXiv 2502.05171; HF: tomg-group-umd/huginn-0125 (Apache-2.0); code: github.com/seal-rg/recurrent-pretraining
- Park et al., LoopUS: Recasting Pretrained LLMs into Looped Latent Refinement Models, arXiv 2605.11011; code: github.com/thrillcrazyer/LoopUS; HF: Thrillcrazyer/Qwen3_1.7B_LoopUS
- McLeish et al., Teaching Pretrained Language Models to Think Deeper with Retrofitted Recurrence, arXiv 2511.07384; code: github.com/mcleish7/retrofitting-recurrence
- Zhu et al., Scaling Latent Reasoning via Looped Language Models (Ouro), arXiv 2510.25741; HF: ByteDance/Ouro-1.4B (Apache-2.0)
- ARLoop, arXiv 2611.05234; HF: osunlp/ARLoop-0.6B, ARLoop-1.3B. Skeptical and mechanistic: arXiv 2507.02199 (latent CoT critique), 2506.18233 (looping buys computation, not knowledge), 2610.19768 (loop-retention), 2610.40325 (how to loop), 2611.06206 (BabyLM data-budget result), ICLR 2027 OpenReview vJgG20sjea (weight tying and gradient pooling)
- Structured/time-series and RL evidence: arXiv 2605.11262 (latent CoT on structured data), 2609.36108 (LoopICL), 2610.41819 (ElTab), 2610.19552 (Test-Time Forecasting), 2609.37432 (Looped Actor), 2310.07820 (LLMTime), 2403.07815 (Chronos), 2310.01728 (Time-LLM)

13.2 Jev, System One models and AI-trading evidence

- TypeSafe AI: typesafe.ai blog, Introducing System One Models and Jev (15 Sep 2026); Jev 1.13 failure-modes documentation; LangChain integration post

(langchain.com/blog/building-a-harness-with-jev); Wikipedia, Jev (AI model); TechCrunch coverage (18 Sep 2026)

- Verified by direct full-text read: the 17-day Jev trading experiment, Towards AI (19 Oct 2026); Miles Deutscher, The Most Powerful AI Trading Bot Possible (X article, 29 Sep 2026). Competitors: OpenAI Decisions (DevDay, 6 Oct 2026), Cloudflare Clef (open weights, Apache-2.0), Perplexity Decisions; community trackers madewithjev.com
- AI-trading evidence: TradingAgents arXiv 2412.20138 + repo history; Alpha Illusion arXiv 2605.16895; StockBench arXiv 2510.02209; FINSABER arXiv 2505.07078; TSFM blind spots arXiv 2610.02058 and 2607.04919; LLM-for-TS ablations arXiv 2406.16964; tabular GBDT evidence arXiv 2207.08815 and 2305.02997; Lopez-Lira and Tang arXiv 2304.07619; Red Queen's Trap arXiv 2512.15732

13.3 Individuals, statistics and risk

- Kaggle: Jane Street 2024-25 final leaderboard and discussion (2nd-place reproduction: kaggle.com/discussion/742687, github.com/cweill/jane-street-causal-forecasting); Optiver TAC 1st-place writeup (hyd); Jane Street 2021 1st-place writeup. Numerai leaderboard (numer.ai/leaderboard/data-scientists, live 6 Nov 2026). Rob Carver live performance 2014-2023 (qoppac.blogspot.com, fetched). Day-trader studies: Barber et al. (JFM 2014); Chague et al. (SSRN 3423101). Retail loss statistics: FCA CP16/40; AMF; OANDA site disclosure
- Validation and risk methodology: Bailey and Lopez de Prado (DSR, SSRN 2460551; PBO, Journal of Computational Finance 2016; PSR, SSRN 1821643); Advances in Financial Machine Learning (2018); Guo et al. arXiv 1706.04599 (calibration); GPT-4 Technical Report arXiv 2303.08774 (RLHF calibration loss). Tooling: purgedcv, ml4t-diagnostic, deflated-sharpe, pypbo, vectorbt 1.2.0, quantstats 0.0.86, NautilusTrader 1.231.0

13.4 Market data and infrastructure

- Gold market: World Gold Council Gold Demand Trends Q3 2026 (gold.org); COMEX GC=F daily/1h data (computed 6 Nov 2026); gold-api.com spot quote. Data sources: dukascopy-node v1.51.1 (dukascopy-node.app/instrument/xauusd), histdata.com coverage page, yfinance 1.8.0 (live-tested intraday limits; XAUUSD=X delisting), FRED series DGS10 / VIXCLS / DCOILWTICO / DEXUSEU / DTWEXBGS / SP500 / GOLDPMGBD228NLBM, federalreserve.gov FOMC calendar (verified), BLS release schedules, OANDA v20 practice API docs

- Kaggle and training: kaggle.com/docs (GPU quota), Kaggle staff discussion 361104 (T4x2 transition, Sep 2026), Unsloth requirements and HF T4 benchmarks (huggingface.co/blog/unsloth-trl), flash-attention README (Turing limitations), DDP benchmarks (github.com/nmquan22/multi-gpu-accelerate-vs-lightning)

13.5 Low-confidence items (flagged, not relied upon)

- Reddit r/hermesagent JEV trading-stack post: network-blocked during research; carried as context only. Jev's internal architecture and RLCD details: unpublished; the Brier-loss characterization is Wikipedia-editor inference (MEDIUM-LOW).
- Stooq long-history coverage and OANDA practice-account candle depth for XAUUSD: not verifiable from this environment; both are audited in Phase 1 before reliance. Exact 2026 broker swap rates: derived from rate math, to be confirmed against the target broker's table before live trading.
- Reported TypeSafe AI late-2026 funding round (MEDIUM): from secondary press coverage only. Qwen3.5-0.8B quality figures: from the Artificial Analysis small-models writeup (MEDIUM); its Unsloth 4-bit fine-tuning support is expected but was not directly verified - one of Phase 0's smoke tests.